

Projects List for EE-568 Reinforcement Learning @ EPFL

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1 Theory Projects

Below you can find 16 possible project directions for the *theory track*. Pick **one** of them (e.g., "Policy Gradient") and read the 2–3 papers listed for it. The steps of a project should be the following:

- Quickly scroll through the papers to understand the general idea. Skip the proofs and the appendix for the moment, and try to understand what problem each of the papers is solving.
- Understand how the results of the papers we suggested compare with each other. (For example, they might consider different setups, or they might build on top of each other.)
- Understand whether some of the papers have limitations that other papers in the list solve.
- Understand how the papers could be improved. For example try to think which research questions remains open or seem within reach.
- Write a report of 6–8 pages answering the above questions. Please do include technical writing (i.e., include relevant citations, theorem statements, definitions, etc.).

Theory Project 1: Theoretical improvements on DPO

Papers: Huang et al. [16], Xie et al. [43]

Theory Project 2: Theoretically understanding of GRPO

Papers: Davis and Recht [5], Tajwar et al. [36]

Theory Project 3: Theory of LLM/robotics pretraining

Papers: Chen et al. [4], Wagenmaker et al. [41]

Theory Project 4: Policy Gradient

Papers: Agarwal et al. [1], Mei et al. [26]

Theory Project 5: Online Learning in Bandits

Papers: Lattimore and Szepesvári [23][Chapters 6,7,11]

Theory Project 6: Online Learning in Adversarial MDP

Papers: Even-Dar et al. [10], Zimin and Neu [45], Jin et al. [18]

Theory Project 7: Reinforcement Learning with Linear Function Approximation

Papers: Jin et al. [17], Cai et al. [2], Wang et al. [42]

Theory Project 8: Offline Reinforcement Learning

Papers: Gabbianelli et al. [13], Hong and Tewari [15]

Theory Project 9: Offline Imitation Learning

Papers: Agarwal et al. [1][Chapter 15.1, 15.2, 15.3], Foster et al. [11], Rajaraman et al. [28]

Theory Project 10: Offline Imitation without expert realizability

Papers: Rohatgi et al. [30], Joshi et al. [20], Espinosa-Dice et al. [9]

Theory Project 11: Query-Based Imitation Learning

Papers: Ross and Bagnell [33], Ross et al. [32], Rajaraman et al. [29]

Theory Project 12: Dataset-based Imitation Learning

Papers: Shani et al. [35], Xu et al. [44], Viano et al. [39], Viel et al. [40]

Theory Project 13: Identifiability in Inverse Reinforcement Learning

Papers: Cao et al. [3], Kim et al. [21], Rolland et al. [31]

Theory Project 14: Multi-Agent Reinforcement Learning

Papers: Liu et al. [25], Jin et al. [19], Leonardos et al. [24]

Theory Project 15: Reinforcement Learning in Constrained MDP

Papers: Efroni et al. [8], Ding et al. [7], Vaswani et al. [38]

Theory Project 16: Robust Reinforcement Learning

Papers: Derman et al. [6], Tessler et al. [37], Kumar et al. [22]

2 Applied Project 1: Deep RL

For the *practical track* you are required to re-implement four famous RL algorithms listed below. You should also test them on some of the following OpenAI-Gym environments: `Cartpole`, `MountainCar`, `MountainCarContinuous`, `Acrobot` and `Pendulum`.

In your report of 6–8 pages you are required to compare the **following 4 algorithms** based on of your empirical observation. This means providing appropriate plots and score statistics of your algorithms based on fair comparison between them.

Possible algorithms:

- DQN by Mnih et al. [27]
- PPO by Schulman et al. [34]
- SAC by Haarnoja et al. [14]
- TD3 by Fujimoto et al. [12]

You can also add a qualitative discussion about the algorithms building around the following questions:

- Which algorithm is more computationally expensive per iteration ?
- Which algorithm store the policy more compactly ?
- Which one scales better for continuous actions ?
- Which algorithm makes efficient use of off-policy data ?

Finally, view the report as diary in which you can keep track of the observations made during the implementation process. We are interested in knowing which small details in the implementation you found are crucial to make the algorithm work in practice! For example, if you had a bug that took you a long time fix, write it down. If you found that the algorithm's performance is very sensitive to certain hyperparameter tuning, write it down. Take also note if you find out that an hyperparameter affects the performance only minimally, and think about possible reasons. Corroborate your claims by showing plots that compare the algorithms when run for the different hyperparameters (i.e., do not only report the final, good hyperparameter choices that made it work eventually).

Important: Each plot you present should report an algorithm's performance averaged across at least 3 seeds.

3 Applied Project 2: RLHF

For the *practical track* you are required to test different RLHF algorithms on some of the following OpenAI-Gym environments: `Cartpole`, `MountainCar`, `MountainCarContinuous`, `Acrobot` and `Pendulum`. You need to proceed as follow.

- **Trajectory generation:** Use an RL algorithm of your choice to train a good policy π_1 that achieve quite reliably the highest reward in the environment. Also save a checkpoint during training for a policy π_2 which achieves a reward more or less equal to half the maximum attainable total reward. At this point, generate the preference dataset generating K pairs of preferred and rejected trajectory. To generate each pair, generate one trajectory with π_1 (denoted by τ_1) and one with π_2 (denoted by τ_2). Letting $R(\tau)$ being the total reward of the trajectory τ , let τ_1 be the preferred trajectory with probability

$$\frac{\exp(R(\tau_1))}{\exp(R(\tau_1)) + \exp(R(\tau_2))}$$

- **Run RLHF algorithms** Compare DPO and PPO-RHLF for different sizes of the preference dataset and for at least two of the aforementioned environments

Important: Each plot you present should report an algorithm's performance averaged across at least 3 seeds.

4 Applied Project 3: Imitation Learning

For the practical track you are required to test different imitation learning algorithms on some of the following OpenAI-Gym environments: `Cartpole`, `MountainCar`, `MountainCarContinuous`, `Acrobot` and `Pendulum`. You need to proceed as follow.

- **Trajectory generation:** Use an RL algorithm of your choice to train a good policy π_1 . At this point, generate the expert dataset generating K trajectories rolling out π_1 .
- **Run imitation learning algorithms:** Compare IQ-Learn, one algorithm among (CSIL, HyPE, f-IRL, ML-IRL) and its SOAR enhanced version for different sizes of the expert dataset and for at least two of the aforementioned environments.

Important: Each plot you present should report an algorithm's performance averaged across at least 3 seeds.

5 Interdisciplinary projects

Some other labs at EPFL offer *interdisciplinary projects* that target the application of RL algorithms to other scientific problems. We collected a list of labs that are open towards supervising you on such a project.

If you choose this option you can reach out to **one** of the labs in [this list](#) and express your interest in the project. You are expected to hand in a report of 6–8 pages. It will be evaluated in collaboration with the external lab that hosted your project.

References

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